

LARS SCHÖBITZ

I am developing data science training and provide consulting services to manage the world's poop. My work experience includes every sector under the sun, which has formed me into a highly adaptable and empathetic individual. I like to crochet, bake, cook, run, code, and play in my free time. I am a 101% type of person.

View this CV online with links at lse.de/cv

ME

- now**
|
2012
Loving life in Zurich
Zurich, Switzerland
• Just Switzerland
📍 Wollyhood
- 2020**
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2011
Multi-cultural work experience
Mother Earth
• South Africa, Uganda, Vietnam, India, Tanzania, Haiti, USA, Bhutan, Senegal
📍 Global
- 2018**
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2017
Loved life in London
London, UK
• So much fun
📍 Brixton
- 2012**
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2011
Sanitation enthusiast
Durban, South Africa
• Living with, and learning from Prof. CA Buckley
📍 49 Essex Grove
- 1986**
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1986
German citizen
Haemelhausen, Germany
• Raised in a town of 600 people, I grew up with two generations, a large vegetable and fruit garden, animals and a christmas tree farm.
📍 Flat plains of Lower Saxony

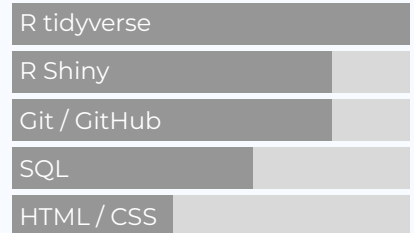
EDUCATION

- 2020**
|
2019
RStudio Instructor Programme
www
• Certified as an RStudio Instructor⁷
📍 RStudio
- 2012**
|
2008
Environmental and Hygiene Engineering (Dipl. Ing.)
Giessen, Germany
• Thesis²: Performance Evaluation of a full-scale DEWATS plant in South Africa
📍 Mittelhessen University of Applied Sciences

CONTACT

✉ lars@lse.de
🐦 larnsce
🐙 github.com/larnsce
🔗 lse.de
in [LinkedIn](#)

LANGUAGE SKILLS



Made with the R package [pagedown](#).

The source code is available on github.com/larnsce/cv.

Last updated on 2022-02-14.

- Scholarship with ERASMUS (European Exchange Program)**
 Umea, Sweden  Umea University
 2010
 |
 2010
 • Main subject: Northern Aquatic Systems, including sampling of different surface water and analysis of DOC, COD, main nutrients, heavy metals
- Change and Project Management**
 Mikkeli, Finland  Mikkeli University
 2009
 |
 2009
- General qualification for university entrance**
 Verden, Germany  Gymnasium am Wall
 2006
 |
 2003



WORK EXPERIENCE

- Open Science Specialist**
 Zurich, Switzerland  ETH Zurich
 now
 |
 2021
 • Support faculty, PhD, and MSc. students in applying reproducible workflows in their research.
 • Oversee aspects of reproducibility and open science principles for published journal papers
 • Manage online and social media channels
 • Setup organisational work, task, and project management tools
- Data Science Instructor**
 Zurich, Switzerland  Statistical Office of the Canton of Zurich
 now
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 2021
 • Teaching a data science with R course to civil servants
 • Course website: <https://rstatsz.github.io/website/>
- Data Science Instructor**
 Boulder, USA  Colorado University
 now
 |
 2021
 • Developing and teaching an online module to teach R
- Director, Founder**
 Zurich, Switzerland  Lars Schoebitz GmbH
 now
 |
 2018
 • Teaching Data Science with R
 • Consultant for city-wide faecal sludge management systems
 • Supporting organisations in establishing strong monitoring and evaluation frameworks
- Data and Impact Expert**
 Essex, UK  Aquaconsult Limited
 2021
 |
 2018
 • Venture Advisor for Transition to Scale Program in Urban Sanitation by Grand Challenges Canada
 • Support to Clean Team, Ghana; Eau&Vie, Bangladesh; Safi Sana, Ghana

I believe in the opportunity of the human rights based approach to water and sanitation above the belief in a billion dollar sanitation economy.

2020
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2020

● **Data Science Intern**

Zurich, Switzerland

📍 Statistical Office of the Canton of Zurich

- Developed R Shiny Dashboard³ for analytics of incoming data analysis requests
- Developed R package for web analytics with Matomo⁴
- Published fully reproducible news article⁵
- Supported activities on COVID19 monitoring⁶ and data analysis
- Organised a session at DINAcon 2020⁷
- Responded to data analysis requests from civil society, politics, universities, private and public sector

2020
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2020

● **Consultant**

Duebendorf, Switzerland

📍 Eawag/Sandec

- Supported the preparation of factsheets for a publication on city-wide inclusive sanitation (CWIS) approaches and tools

2020
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2017

● **Data Scientist**

Putnam, USA

📍 Biomass Controls PBC

- Building tools for data visualisation and business intelligence using R Shiny framework
- Support of R&D work and preparation of results for publications and conferences

2020
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2019

● **Lead Consultant**

Cap-Haitien, Haiti

📍 Inter-American Development Bank

- Team of Consultants: Berta Moya, Anthony Kilbride, Andy Peal
- Development of city-wide faecal sludge management systems
- Detailed review of design proposal for faecal sludge treatment plant
- Preparation of shit-flow-diagram (SFD) for Cap-Haitien

2018
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2018

● **Data Scientist**

London, UK

📍 Gather

- Supported conceptualization of data dive for urban sanitation data
- Cleaned and prepared toilet census data set of Kanyama compound (Lusaka, Zambia) for analysis

2018
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2016

● **Consultant**

Geneva, Switzerland

📍 World Health Organization

- Global Monitoring of the Sanitation Chain: Proof of concept testing for monitoring indicators on safely managed sanitation (Target 6.2) and safely treated wastewater (Target 6.3) in the SDG agenda
- Production of country files for SDG 6.3.1 Proportion of wastewater safely treated

2016
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2012



Research Project Manager

Duebendorf, Switzerland

📍 Eawag/Sandec

- Project management with global, multi-disciplinary teams and topics of wide spread interest (RRR Business Models, SFDs Methods and Tools development)
- Research lead on several projects that aim to develop methods for diagnostic tools in faecal sludge management
- Mentor, lecturer and effective networker

2012
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2011



Research Assistant

Durban, South Africa

📍 PRG, University of KwaZulu-Natal

- Drove a 1950s Land Rover on Durban highways to collect water samples from DEWATS plant
- Emptied pit latrines with spades and shovels
- Built a snooker-like table to separate pit contents into their individual constituents



TEACHING EXPERIENCE

2021
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2021



R Instructor

www

📍 Statistical Office of the Canton of Zurich

- Developing a training for Data Science with R for⁸ civil servants

2017
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2017



Lecturer

Giessen, Germany

📍 Mittelhessen University of Applied Sciences

- Introduction to Faecal Sludge Management⁹ for environmental engineering students

2016
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2013



Lecturer

Switzerland

📍 Lecturer NADEL, ETH Zurich, EPF Lausanne

- Teaching on basics of Faecal Sludge Management

2016
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2016



Lecturer

Switzerland

📍 Massive Open Online Course: Planning & Design of Sanitation Systems and Technologies

- Prepared module on Shit Flow Diagrams¹⁰



PRO BONO

now
|
2018



Expert Advisor

London, UK

📍 Gather

- Providing support with my passion for open data science and by sharing knowledge on urban sanitation

I am passionate about teaching coding with R to novel users.

2020
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2017



R Instructor

Zurich, Switzerland

- Teaching employees of Eawag/Sandec in using R for statistical computing software
- Started out with lectures in 2017⁷⁷ and continued individual one-to-one support to date

2020
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2018



Technical Expert

Global

📍 ISO Project Committee 318

- Member of the US Technical Advisory Group for emerging standard ISO/PC318 “Community scale resource oriented sanitation treatment systems”



TEACHING EXPERIENCE



PUBLICATIONS

2020



Corona veraendert Alltagssorgen nur wenig¹¹

📍 Web article

- Published source code and data as GitHub repository¹²

2020



Hotel-Tourismus in Zuerich – Geschaefsttouristen und auslaendische Gaeste hinterlassen Luecke¹³

2020



Estimating Safely Managed Sanitation in Urban Areas; Lessons Learned From a Global Implementation of Excreta-Flow Diagrams¹⁴

📍 Frontiers in Environmental Science

2019



City Service Delivery Assessment for Citywide Inclusive Sanitation¹⁵

📍 [Inclusive Sanitation in Practice](#)

- Reviewer

2019



Towards an off-grid fecal sludge treatment unit: demonstrating energy positive thermal treatment¹⁶

Gates Open Research

- Prepared fully reproducible research paper¹⁷

2018



Methods to reliably estimate faecal sludge quantities and qualities for the design of treatment technologies and management solutions¹⁸

📍 Journal of Environmental Management, 223, 898-907

- Collection and laboratory analysis of 180 faecal sludge samples in Kampala, Uganda
- Data analysis using R

I support open science principles.

- 2018 ● **Remote Sensing, Mobile Applications and Open Data Science Tools for Better Monitoring of Sanitation Systems¹⁹**
 📍 In HIC 2018. 13th International Conference on Hydroinformatics
 • Conference paper
- 2018 ● **Resource recovery from waste: business models for energy, nutrient and water reuse in low- and middle-income countries²⁰**
 📍 Oxon, UK: Routledge; Earthscan. 816p
- 2018 ● **Sustainable Development Goal 6 Synthesis Report 2018 on Water and Sanitation²¹**
 📍 United Nations
- 2018 ● **Progress on safe treatment and use of wastewater: piloting the monitoring methodology and initial findings for SDG indicator 6.3.1²²**
 📍 World Health Organization and UN-HABITAT
- 2018 ● **Country files for SDG 6.3.1 Proportion of wastewater safely treated²³**
 📍 World Health Organization
- 2018 ● **Guidelines on sanitation and health²⁴**
 📍 World Health Organization
- 2017 ● **GIS analysis and optimisation of faecal sludge logistics at city-wide scale in Kampala, Uganda²⁵**
 📍 Sustainability, 9(2), 194
- 2017 ● **GIS analysis and optimization of faecal sludge logistics at city-wide scale in Kampala, Uganda. Supplementary Report²⁶**
- 2017 ● **Market driven approach for faecal sludge treatment products²⁷**
 📍 In WEDC conference papers. WEDC conference 40 (p. 2807 (6 pp.))
- 2014 ● **FAQ: faecal sludge quantification and characterization; field trial of methodology in Hanoi, Vietnam²⁸**



TECHNICAL REPORTS

I strive for efficiency and approach life with 'All or Nothing'.

2016
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2015

● Sanitation service delivery context analysis and mapping of excreta flows along the sanitation service chain and throughout the city

- Download Bignona/Senegal¹¹
- Download Danang/Vietnam¹²
- Download Dar es Salaam/Tanzania¹³
- Download Durban/South Africa¹⁴
- Download Hanoi/Vietnam¹⁵
- Download Kampala/Uganda¹⁶
- Download Khulna/Bangladesh¹⁷
- Download Nonthaburi/Thailand¹⁸
- Download Tikapur/Nepal¹⁹



SANDEC NEWS ARTICLE

2016
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2013

● Sandec News Articles

- Illustrating the Fate of Excreta: The SFD Methodology in Eight Cities¹¹
- Private Public Partnership Solutions for Faecal Sludge Management¹²
- FAQ: Faecal Sludge Quantification and Characterization; Kampala¹³
- Waste-based Business Models for Resource Recovery¹⁴

I have a mindset of learning from failures and am a signatory of the [Nakuru Accord](#)



CONFERENCES & SEMINARS

2020

● UN World Data Forum 2020

www

2020

● DINacon

www

- Session organiser: Unsere Antwort auf COVID19: Ein Beispiel von Kollaboration und Gemeinschaft im Open Data Zeitalter¹¹

2019

● Latinosan Conference

San Jose, Costa Rica

- Invited presenter and moderator of session¹² on sludge management organized by the Inter-American Development Bank
- Presentation Slides¹³

2019

● 5th Faecal Sludge Management Conference

Cape Town, South Africa

- Oral Presentation¹⁴ on Development of Tools for Efficient Remote Monitoring of Faecal Sludge Treatment Units
- Poster Presentation on Combined Heat and Power with Micro Organic Rankine Cycle for Off-Grid Sanitation Treatment
- Invited presenter at side meeting organised by Centre for Science and Environment India. Development of a Faecal Sludge Data Repository¹⁵

I am optimistic, direct, patient and empathetic.

- 2019 ● **WaterLex, 1st World Summit on Leaving No One Behind**
 Geneva, Switzerland

 - Finalist for project on human-rights based solutions to water and sanitation. Testing innovative technologies for Menstrual Hygiene Management in humanitarian and public settings
 - Oral Presentation¹⁶

- 2018 ● **2nd meeting of ISO PC 318 WG 1**
 Toulouse, France

 - Technical Expert for emerging standard representing the US technical advisory group

- 2018 ● **Monitoring, Evidence, and Dissemination for Scale (MEDS) convening**
 Dakar, Senegal

 - Hosted by the Bill & Melinda Gates Foundation (BMGF)

- 2018 ● **13th International Hydroinformatics Conference**
 Palermo, Italy

 - Oral presentation¹⁷ on Remote sensing, mobile applications and open data science tools for better monitoring of sanitation systems

- 2018 ● **Stakeholder meeting for FSM laboratory of CSE at Anil Agarwal Environment Training Institute (AAETI)**
 Tijara, India 📍 CSE AAETI

 - Invited presenter

- 2018 ● **Colorado WASH Symposium**
 Boulder, USA 📍 University of Colorado

 - Oral presentation¹⁸ on Development of tools for efficient remote monitoring of faecal sludge treatment units

- 2018 ● **Gather Hub Data Dive**
 London, UK

 - Keynote speaker Gather Hub's data dive¹⁹ on urban sanitation for Lusaka, Zambia
 - Presentation Slides²⁰

- 2017 ● **4th Faecal Sludge Management Conference**
 Chennai, India

 - Monitoring Safely Managed Sanitation on the 2030 Agenda for Sustainable Development; Experience from Uganda
 - Presentation²¹
 - Abstract²²

- 2016 ● **UNC Water and Health Conference**
 Chapel Hill, USA

 - Safely-Managed Sanitation: Monitoring higher levels of service in SDG Target 6.2
 - Event support <https://kutt.it/p4toLj>

- 2016 ● **Sandec Research Seminar**
 Duebendorf, Switzerland

 - Shit-Flow-Diagrams and the potential for monitoring of Sustainable Development Goals

- 2015 ● **UNC Water and Health Conference**
 Chapel Hill, USA

 - City level excreta flow analysis; the SFD Promotion Initiative
 - Poster²³

- 2015 ● **Stockholm World Water Week, Sweden**
 Stockholm, Sweden

 - Side Event²⁴: Liveable Cities for All: Integrated Sanitation and Water Services
 - Side Event²⁵: SFD, a tool to foster sustainable urban sanitation programming
 - Sustainable City Sanitation; from planning to implementation. Event support²⁶

- 2015 ● **4th AfricaSan Conference**
 Dakar, Senegal

 - Side Event: SFD a tool to boost urban sanitation. Presentation Slides²⁷
 - Panellist: Public-Private Partnerships to Meet the Sanitation Shortfall

- 2015 ● **3rd Faecal Sludge Management Conference**
 Hanoi, Vietnam

 - Quantification of FS by truck counting studies with the Private Emptiers Association in Kampala Abstract pp. 6-8²⁸, Presentation²⁹
 - Scaling-up sanitation businesses in low- and middle-income countries Presentation³⁰
 - FSM / Treatment processes for resource recovery / Session 1: Moderation³¹
 - Workshop Preparation, Facilitation³²: FSM Systems Approach for Implementation and Operation: Research into Practice: Towards standardization of FS quantification and characterization on a city-wide scale

- 2014 ● **37th WEDC International Conference**
 Hanoi, Vietnam

 - FAQ: Faecal Sludge Quantification and Characterization: field trial of methodology in Hanoi, Vietnam, Conference Paper³³

- 2014 ● **Stockholm World Water Week, Sweden**
Stockholm, Sweden
• Side Event: From Faecal Sludge to Fuel: Safe Sanitation with Business Opportunities
- 2013 ● **36th WEDC International Conference**
Nakuru, Kenya
• Espresso Shot Presentation on Resource, Recovery & Reuse
- 2012 ● **2nd Faecal Sludge Management Conference**
Durban, South Africa
• Business model innovations for scaling-up faecal sludge businesses in developing countries: Abstract³⁴ Presentation³⁵
- 2012 ● **Water Institute of Southern Africa (WISA) conference**
Cape Town, South Africa
• Practical experience and operating data from a full-scale decentralised wastewater treatment plant treating domestic wastewater



REFERENCES

- now ● **Dr. Christian Zurbrugg¹¹**
Duebendorf, Switzerland  Eawag/Sandec
• Member of the Directorate & Group Leader Solid Waste Management
- now ● **Prof. CA Buckley¹²**
Durban, South Africa  PRG, University of KwaZulu-Natal
• Co-Head
- now ● **Dr. Kate Medlicott¹³**
Geneva, Switzerland  World Health Organization
• Team Leader Sanitation